CIRCLE × ARC HACKATHON 2026

ASM

Agent Service Manifest — The first protocol that lets AI agents discover, evaluate, and pay for services autonomously. Per API call. Sub-cent precision. Settled on Arc in USDC.

0	$0.25	15	$0.00
N TX	USDC VOLUME	AI SERVICES	PER CALL

Track: Per-API Monetization Engine

Yi Guo · April 2026 · MIT License

THE PROBLEM

Agents Have Zero Data to Choose Between Service Providers

When an AI agent needs to call an API — translate text, generate an image, transcribe audio — it faces a choice between dozens of providers. Today it has no structured data to make that decision.

WITHOUT ASM

- Blind selection via hardcoded API keys
- 3–10x cost overrun or quality mismatch
- Decisions are non-reproducible and unexplainable
- Zero intelligence at the selection step
- No price discovery mechanism exists

WITH ASM

- Structured multi-criteria matching across 47 categories
- Optimal cost-quality tradeoff via TOPSIS ranking
- Deterministic, auditable, fully explainable decisions
- Full autonomous decision capability per call
- Sub-cent payment settles automatically on-chain

This is not a model intelligence problem — it's a data problem.

WHY NOW

The Agent Economy Is Exploding — But Infrastructure Is Missing

> "By 2028, AI agents will orchestrate over $4T in API transactions annually. Every one of those calls is a pricing decision that today happens blindly."

Agent Proliferation

MCP, A2A, LangChain, CrewAI — agent frameworks are multiplying. Each agent needs to choose between 10-100+ service providers per task category.

Agent frameworks (2024→2026) +340%

API Spend Growth

Enterprise AI API spending grew from $15B (2023) to $58B (2025). Most of that spend is sub-optimally allocated because there's no comparison layer.

AI API market CAGR ~97%

The Gap

MCP solves "what can a tool do." A2A solves "how agents talk." AP2/x402 solves "how to pay." Nobody solves **"what is a service worth."**

Value-discovery coverage 0%

THE SOLUTION

Three Layers That Turn Any Agent Into an Autonomous Service Buyer

01 Discover

Structured service manifests across 6 categories and 47 taxonomy nodes. 70 real-world services registered with pricing, quality metrics, SLA, and payment addresses.

LLM | Image | Video | TTS | Embedding | GPU

02 Evaluate

TOPSIS multi-criteria decision engine with configurable preference weights and hard constraints. Python + TypeScript parity verified. Discovery accuracy: **72% top-1 match**.

TOPSIS | Weighted Avg | Trust Delta

03 Pay

Circle Nanopayments on Arc via x402 protocol. $0.002–$0.005 USDC per API call. Fully autonomous settlement — no manual wallet wiring. One endpoint, N recipients.

x402 | USDC | Arc

Agent Task → Taxonomy Match → Registry Query → TOPSIS Score

→ Service Call → USDC Receipt → Trust Update

HOW IT WORKS

One Request. Full Autonomous Loop.

1. **Agent Sends Natural Language Task**

   "Translate this document to Japanese using the best available LLM"

2. **LangGraph Discovers Taxonomy**

   AI-powered embedding → retrieval → reranking maps request to `ai.nlp.translation`

3. **TOPSIS Ranks All Candidates**

   Evaluates cost, quality, latency, io_ratio against agent preferences → deterministic winner

4. **x402 Settles Payment on Arc**

   $0.005 USDC sent to provider's wallet via Circle Gateway batching protocol

5. **Signed Receipt Updates Trust**

   Cryptographic proof of actual performance feeds back into trust_delta score

EXAMPLE: POST /API/SCORE

```
const result = await
fetch('/api/score', {
  method: 'POST',
  body: JSON.stringify({
    task: 'translate doc to
japanese',
    preferences: {
cost_weight: 0.4,
quality_weight: 0.5 }
  })
});

// Response:
{
  "winner": "deepL-pro",
  "taxonomy":
"ai.nlp.translation",
  "price": 0.003,
  "txHash": "0x7f3c...9a2e",
  "settled": true
}
```

200 OK | $0.003 USDC | Arc Confirmed

WHY TRADITIONAL GAS MODELS FAIL

Micro-Payments Need Micro-Cost Settlement

All submissions must demonstrate why their model would fail with traditional gas costs. Here's ours:

100× cheaper per transaction

Batched Settlement

Multiple micro-payments settled in a single on-chain transaction via Circle Gateway's x402 batching protocol. Gas amortized across hundreds of calls.

USDC-Native Budgeting

No volatile gas token exposure. Agents budget in stablecoins with predictable per-call costs. No surprise spending from gas volatility.

The Math Doesn't Work Otherwise

At $0.50+/tx, paying $0.005 for an API call means gas costs **100× the payment itself**. The entire model collapses without nanopayments.

LIVE DEMO RESULTS

Real Transactions. Real Data. No Simulations.

Every number below comes from actual on-chain transactions executed during development. All verifiable on Arc Block Explorer.

50
ON-CHAIN TRANSACTIONS

$0.25
TOTAL USDC VOLUME

15
UNIQUE AI SERVICES

6
AGENT PERSONAS TESTED

AGENT PERSONAS × SCENARIOS

AGENT	TAXONOMY	STRATEGY
ChatBot Agent	`ai.llm.chat`	Cost-optimized LLM
Creative Agent	`ai.vision.*`	Quality-first image gen
Voice Agent	`ai.audio.tts`	Natural speech TTS
RAG Agent	`ai.llm+embedding`	High io_ratio

HACKATHON REQUIREMENTS CHECKLIST

Per-action pricing ⩽ $0.01	PASS · $0.005
50+ on-chain transactions	PASS · 50/50
Margin explanation	PASS · See slide 6
Uses Circle + Arc + USDC	PASS · All three
Transaction video proof	REQUIRED · Video

AGENT	TAXONOMY	STRATEGY
		optimization
DevOps Agent	`cloud.compute.gpu`	Speed + reliability
Multi-Modal Agent	`all categories`	Balanced cross-category

ON-CHAIN PROOF LINKS

Buyer: `0xF5d4...b038`

Gateway: `0x0077...19B9`

TRUST MODEL

Verify, Don't Trust — 3-Layer Trust Architecture

From self-reported claims to cryptographic proof. Aligned with ERC-8004 computable trust dimensions.

Transparency Layer

Providers submit structured manifests with self-reported metrics, pricing dimensions, and transparency flags. Every claim is machine-readable and queryable.

Verification Layer

Independent benchmarks and reputation scores attached with source URLs. Third-party audit data cross-referenced against declared metrics.

Receipt Layer

Signed receipts (IETF ACTA draft) cryptographically prove actual performance per API call. Post-hoc verification creates a self-sustaining trust loop.

$$\text{trust_delta} = |\text{declared} - \text{actual}| \,/\, \text{declared}$$

Each transaction updates the provider's credibility score. Payment → Delivery → Verification loop sustains itself.

WHERE ASM FITS

The Missing Value Layer in the Agent Economy Stack

ASM doesn't replace existing protocols — it completes them by adding the economic dimension nobody else covers.

Model Context Protocol
"What a tool CAN DO" — capability discovery
SOLVED

Agent-to-Agent Protocol
"How agents COMMUNICATE" — message passing
SOLVED

Agent Payment Protocol (x402)
"How to PAY safely" — settlement layer
SOLVED

Agent Service Manifest
"What a service IS WORTH" — value discovery & evaluation
NEW — THIS PROJECT

THE COMPLETE STACK

MCP discovers capability → **ASM** evaluates value → **AP2/x402** settles payment. Together they form the complete autonomous agent economy stack. Without ASM, agents can find tools and pay for them, but they cannot make intelligent economic decisions about which tools are worth paying for.

ROADMAP

From Prototype to Protocol

PHASE 0 — NOW

April 2026 · Hackathon

Working Prototype

70 manifests across 47 taxonomies · TOPSIS scorer (Py+TS) · MCP Server (5 tools) · x402 integration · 50/50 test transactions on Arc · Gemini function calling

PHASE 1 — Q2 2026

Mainnet Deployment

Production Readiness

Automated manifest crawler · Provider self-service portal · LangChain SDK · Mainnet settlement · Expanded benchmark suite

PHASE 2 — Q3 2026

Network Effects

100+ Services & Multi-Chain

100+ registered services · Trust reputation marketplace · Multi-chain support (Base, Arbitrum, Arc) · Community governance

PHASE 3 — Q4 2026

Protocol Maturity

Full Decentralization

Agent SDK (Python, TypeScript, Rust) · Decentralized registry · Governance token · ERC-8004 full compliance

TECH STACK SUMMARY

Settlement	Arc Testnet + USDC + Circle Gateway x402
Scoring	TOPSIS + Weighted Average + Trust Delta
Discovery	LangGraph state machine + Embedding pipeline
AI	Gemini FC + AIMLAPI fallback + Rule Engine
Registry	MCP Server (5 tools) + REST API + 70 manifests

OPEN SOURCE

MIT License. All code, manifests, and scoring logic publicly available. Designed as an open protocol — not a walled garden.

Open Source · MIT License

Thank You

Agent Service Manifest × Circle Nanopayments

The value layer for the autonomous agent economy.

View Source Code

Live Demo App

github.com/calebguo007/asm-arc-circle-2026 · @calebguo007 · Yi Guo

Arc | USDC | Circle Gateway | x402 | Nanopayments | TOPSIS | LangGraph | MCP